

Joint-centre offset dominates 3D position error and can reverse comparisons between markerless triangulation methods

Title page

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Abstract

Markerless motion capture pipelines are commonly tuned and compared using three-dimensional joint position error against a marker-based reference. However, pose-estimation keypoints and marker-based joint centres are not the same anatomical construct and have been reported to differ at the hip and knee by several centimetres, so it is unclear what such a comparison measures. We held one pipeline fixed and varied four triangulation decisions (confidence weighting, rejection by reprojection or object-space error, and Gauss–Newton refinement) and a population offset correction, one at a time, across 104 overground walking trials from 11 participants of the BioCV dataset, recorded on a nine-camera ring against a synchronised 15-camera reference. At the hip and knee the keypoint lay 27–37 mm from the reference joint centre, which accounted for 74–81% of squared position error within a participant and 47–70% pooled across participants, against 2.5 mm for the largest

triangulation effect. When each participant's own offset was subtracted, per-joint verdicts changed. Both rejection criteria appeared to improve hip position as measured, but worsened it by 2.0 and 1.7 mm respectively once the offset was removed, leaving the estimate noisier about its own mean yet better centred on the reference. No triangulation effect at the hip, knee or ankle exceeded 2.5 mm and no single-step angle effect exceeded 0.38° on the four chains tested. All contrasts came from one rig, task and laboratory, and the family structure was exploratory. These findings suggest per-joint position verdicts should be read against the joint-centre offset they are likely to contain.

Keywords: markerless motion capture; multi-view triangulation; gait analysis; joint centre; validation methodology

1. Introduction

Markerless motion capture has been reported to agree with marker-based systems to within a few degrees for lower-limb kinematics (Pagnon et al., 2022; Uhlich et al., 2023), although this evidence is concentrated in the sagittal plane (Cheng et al., 2025) and agreement is poorer for rotations about the long axis of a segment, which Wade et al. (2022) place at $6.9\text{--}13.2^\circ$ (see also Varcin and Boocock, 2026). As these systems move towards clinical and field use, the decisions inside the reconstruction pipeline, such as how views are combined and which are discarded, are increasingly tuned and compared.

The metric used for that tuning was inherited from computer vision, whose founding benchmark reports angle error alongside position error (Ionescu et al., 2014), but pipeline decisions are typically checked against position alone. Of four studies that vary such a decision, two evaluate it on position while reporting angles and two report one outcome only; none relates the two (Karashchuk et al., 2021; Needham et al., 2021; Pagnon et al., 2021; Uhlich et al., 2023). Since landmark misplacement propagates to joint kinematics unevenly

across joints and planes (Della Croce et al., 2005), an intervention that displaces an angle's three defining joints by different amounts need not change that angle proportionately. Pose-estimation keypoints and marker-based joint centres are also not the same anatomical construct. Needham et al. (2021) reported differences of 30–50 mm between the two at the hip and knee, and Kanko et al. (2021) independently observed 3.6 cm at the hip against 2.2–2.4 cm at the knee and ankle, varying over the gait cycle. An offset of this kind is not produced by triangulation and is common to both variants compared. Whether a difference in position error between them may therefore be dominated by the shared offset, and not by the decision under test, does not appear to have been examined. We therefore asked how much of the position error at the hip and knee the keypoint-to-joint-centre offset accounts for, and whether comparisons between triangulation variants change when that offset is removed. We held a single pipeline fixed and varied five components one at a time, confidence weighting, rejection by reprojection error, rejection by object-space error (Lu et al., 2000), Gauss–Newton refinement, and a population offset correction. We also asked whether a gain in position transmits to joint angle, testing each claimed dissociation as an interaction, since two separate verdicts do not establish one (Gelman and Stern, 2006).

2. Methods

2.1. Dataset

The BioCV dataset was analysed (University of Bath Research Data Archive, doi:10.15125/BATH-01258; Evans et al., 2024). Eleven of the fifteen healthy volunteers the archive describes were present in our copy. These eleven (six male, five female, aged 26.2 ± 5.1 years) each performed ten straight-line overground walking trials at a self-selected pace, except one who performed nine. Five trials were lost to missing reference data or unusable

video, leaving 104, from each of which 60 near-evenly spaced frames were retained (6,240 frames). At 45 ms between frames no temporal filter was possible (Table S16). Video was recorded by a nine-camera ring at 1920×1080 and 200 Hz. The lenses distort strongly, and correction under the model of Brown (1971) displaces a point by 6.09 px on average and 84.80 px at worst, so all results are conditional on it (Table S2b). Reference kinematics came from a synchronised 15-camera Qualisys Oqus system at 200 Hz, processed with the dataset's own six-degree-of-freedom model. The two systems were hardware-triggered and a whole-frame offset was corrected post hoc from an LED pulse. The providers do not quantify the sub-frame residual, which we bound at typically 3.6 mm at the hip and knee, and their reported alignment error of 3.62 mm was not recomputed (Tables S12, S14). Keypoints were detected with RTMPose (Jiang et al., 2023) and each COCO-17 keypoint was differenced against the identically named output of the reference model, so every offset reported here is the keypoint minus the model joint centre. That model builds joint centres as the midpoint of the medial and lateral markers at every joint except the hip, where it applies the regression of Bell et al. (1989). Two further detectors, RTMO and HALPE-26, were run on the same frames (Table S9).

2.2. Pipeline arms

An arm is a complete pipeline configuration; all arms shared detections, calibration and candidate cameras, differing only in the named component, and contrasts are named with the baseline first. We varied confidence weighting (the detector's confidence scores against uniform weights), the two rejection criteria below, Gauss–Newton refinement against the direct linear transform, and a population offset table, a fixed per-joint translation fitted to this rig and held out in participant only. The distance weightings and matched-retention arms are diagnostic controls, so claims about a deployable rule rest on the adaptive comparison alone

(Table S10). The two estimators minimise Σwr^2 and $\Sigma w^2 r^2$ respectively, and contrasts that hold this fixed are labelled w^2 -matched.

2.3. The two rejection criteria

For camera i the reprojection residual is the norm of the difference between the projected point and the detection, in pixels. The object-space residual is the perpendicular distance in millimetres from the reconstructed point X to the ray through that detection, the collinearity error minimised by Lu et al. (2000). Those authors report their algorithm to be more robust to outliers than Levenberg–Marquardt alternatives, a claim about an estimator rather than about either residual's power to flag a bad observation, so the comparison drawn here is our own. They also note that this error grows with distance from the camera (Table S2a), and for small angular residuals

$$e^{obj} \approx (Z/f) \cdot e^{px} \quad (1)$$

where Z is the depth of X from that camera and f is its focal length in pixels. Object-space error is therefore a depth-scaled reprojection error, and on a given rig the two criteria differ only through the way Z/f is distributed across the cameras.

Both criteria used the same adaptive rule, discarding the camera with the largest residual whenever it exceeded the median plus three scaled median absolute deviations, re-triangulating after each discard, and stopping at a floor of four cameras. In the matched-retention arms a fixed count replaced this rule, so only the ordering of discards differed.

2.4. Outcomes

Position error was measured on two joint sets, a peripheral set of both ankles and both wrists, labelled [pos:AW] in the tables, and a lower-limb set of the hip, knee and ankle of each side, labelled [pos:HKA]. The latter was accumulated inside the angle computation, sharing frames and arms with the angle outcomes.

The angle outcomes were three-point included angles and no kinematic model was fitted. They were computed identically for the estimate and the reference and named for the segments defining them: the thigh–shank angle, angle(hip, knee, ankle), which approximates knee flexion; its frontal-plane form; and the trunk–thigh angle, angle(mid-shoulder, hip, knee), which approximates hip flexion but also moves with trunk lean. The lower-limb position set contains all three points of the thigh–shank angle but only some of the other two, so a position outcome is matched to an angle's defining joints for that angle only. Four chains are computable under COCO-17, the three angles above and the elbow. Where a chain's terminal keypoint admits more than one choice a pre-declared veto withholds the verdict unless both choices agree in sign (Table S13).

2.5. Statistics

The participant was the inferential unit. Each contrast was reduced to one mean difference per participant and all $2^{11} = 2048$ sign patterns were enumerated, giving an exact two-sided p with a floor of 0.000977 that certifies only that all eleven participants agreed in direction. Multiplicity was controlled with the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) within each of fourteen declared families (Tables S1, S4b), under a positive regression dependence assumed without demonstration (Benjamini and Yekutieli, 2001). Under the arbitrary-dependence modification 39 verdicts fall and are marked † wherever quoted (Table S5), and one correction over the union of the fourteen families moves five verdicts and no headline claim (Table S4). The families were not pre-specified but grew as claims required contrasts, each re-run in full, so the family structure is exploratory (Table S15).

Because significance in one outcome and not in another does not by itself establish a dissociation (Gelman and Stern, 2006), each was tested as an interaction, one relative effect minus the other, computed per participant. The null is equal fractional change, so differences

are also reported in millimetres and degrees, and an effect surviving on only one scale is scale-dependent.

The exact test assumes the two arms are exchangeable within a participant. The shared frames and detections provide that for the triangulation arms but not for the population offset table, whose correction is built from the other ten participants. A rule declared beforehand therefore withholds the q value on every leave-one-out row, leaving a nested null as their only inferential statement (one-sided $p = 0.026$). Every affected family was corrected again without them, and 143 of the 244 contrasts carrying a verdict survived (Tables S3e, S4b, S22). Under leave-one-participant-out refitting the fourteen headline contrasts each retained their direction at $p < 0.05$ (worst $p = 0.029$; Table S6). Intervals invert the sign-flip test, at 95% for a quoted magnitude and 90% for a bounded null (Tables S3k, S8).

3. Results

3.1. The joint-centre offset dominates position error at the hip and knee

Agreement with the reference was 2.575° , 2.637° and 3.651° of mean absolute error on the three angles, and 28.5 mm on the lower-limb position set (Table S2d). The keypoint lay 27–37 mm from the reference joint centre (Table 1). Within a participant that offset accounted for 74–81% of squared position error at the hip and knee and 44% at the ankle. Pooled across participants, as a cohort-level validation study measures it, that scatter moves into the random term and the share falls to 47–70%. The largest triangulation effect at any of these joints was 2.5 mm, an order of magnitude smaller. A difference of norms does not remove a term common to both arms, and the first-order account of what it leaves behind fails at the hip and ankle (Table S7), so the reversals below are empirical and were not anticipated. RTMO and HALPE-26 reproduced the offset on the same frames, and all three detectors disagreed with the reference in the same direction for every participant, although they retain

COCO-17 landmark semantics and so are not independent replications (Tables S9, S13). At the hip the mediolateral component was bilaterally symmetric and directed outward (-18.9 and $+18.5$ mm), a pattern the Bell et al. (1989) regression used by the reference would be expected to produce, whereas a shared timing lag or common-mode registration error would not.

3.2. Removing the offset changes per-joint verdicts

Subtracting each participant's own offset at each joint moved verdicts in both directions (Table 2). Both rejection contrasts reversed at the hip, the object-space criterion moving from $+0.608$ to -1.726 mm and the reprojection criterion from $+0.633$ to -1.983 mm. Because pairing two separate verdicts would not establish a reversal, the difference itself was tested. That gave -2.335 mm on the object-space criterion and -2.616 mm on the reprojection criterion per participant, both at the enumeration floor and surviving correction within a declared family of eighteen ($q = 0.003$). The as-measured halves of those pairs are uncorrected ($p = 0.023$ and 0.044), so the reversal rests on the difference. Two further contrasts reversed at the knee, at matched retention $k = 1$ and $k = 2$. Decomposed by joint, the adaptive criterion improved the knee by 0.339 mm ($q = 0.004$) and seven of twelve per-joint contrasts survived correction (Fig. 1, Table S3b). On both deployable rejection contrasts the hip gained and the knee lost while the three-joint mean was null, so opposed per-joint effects cancelled, although under correction this held only on a matched-retention control. One intervention also survived a comparison between joint subsets, on both scales and under arbitrary dependence (Table S3j). Confidence weighting differed by 1.05 percentage points and 0.276 mm between {hip, knee, ankle} and {hip, ankle} ($q = 0.004$; Table S3n), so the joint set a study summarises over changes what it reports.

3.3. Camera rejection

Against no rejection, adaptive rejection left lower-limb position essentially unchanged, at 0.005 mm worse for reprojection and 0.163 mm better for object-space, bounded within ± 0.35 mm at 90%. That bound was measured on a displaced target. With each participant's own offset subtracted, both criteria worsened hip error about that participant's own mean (Section 3.2; $q = 0.004$). Scatter and centring run in opposite directions. Scatter worsened by 1.801 and 2.082 mm while the centre of the estimate moved 1.648 and 1.958 mm closer to the reference, object-space first in each pair. Within quintiles of the reference angle the same subtraction gave 1.924 and 2.235 mm in that order, so the degradation is not the posture-dependent residual of Fig. 2. The knee and ankle remained null.

The same rejection degraded the thigh–shank angle by 0.188° † and 0.141° and the trunk–thigh angle by 0.192° † and 0.167° †, reprojection first in each pair, which reverses the ordering the corresponding position pairs appear to take ($q = 0.011$ – 0.028 ; Table 3). On the matched-retention control the cost showed a dose-response reaching 0.747° and 0.798° at $k = 3$ (Fig. 3a).

Under Gauss–Newton refinement, reprojection rejection improved lower-limb position by 0.858 mm while degrading the thigh–shank angle by 0.169° ($q = 0.006$ and $0.028^\dagger$), an interaction of 10.04 percentage points ($q = 0.012$; Fig. 4, Table S3h). Of the twenty declared interactions (Table S3i) this is the only one pairing two surviving effects of opposite sign. Its angle component, bounded within -0.016 to $+0.053^\circ$, does not survive the arbitrary-dependence sensitivity, so the dissociation is not demonstrated (Table S11). Gauss–Newton itself cost 1.384 mm against the direct linear transform.

The null appears to be a property of this rig as much as of the rule. On fixed subsets of the same ring both contrasts remained null at seven cameras but broke at five, where discarding bought 0.857 mm on the reprojection criterion and 0.428 mm on the object-space criterion.

One subset per size is not an average, so this bounds the claim rather than measuring a five-camera rig (Table S23). Two controls also cut against the criterion comparison. The object-space advantage fell from 4.62 mm to a non-significant 0.68 mm once lens distortion was corrected (Fig. 3b, Table S3a); and on the elbow chain, outside every declared family and under a subject-cluster bootstrap rather than the exact test, reprojection rejection lowered angle error by 0.295° , the opposite sign to the other chains (Table S3l).

3.4. Confidence weighting and the population offset correction

Confidence weighting improved lower-limb position by 3.10% over uniform weighting (Table 4), the frontal-plane thigh–shank angle by 0.078° ($q = 0.004$) and the trunk–thigh angle by 0.227° ($q = 0.006$). It did not detectably transmit to the thigh–shank angle itself (0.023° , $q = 0.647$) and neither interaction survived, so we claim no angle gain beyond position. The largest single effect was at the peripheral joints, where confidence weighting bought 5.062 mm [+4.309, +5.930]. Propagated through the angle definition the offset alone produces a thigh–shank error of the size observed, though not the same error (Tables S3e, S3f).

Correcting each participant with the mean offset of the other ten improved lower-limb position by 9.848 mm but worsened the thigh–shank angle by 0.606° [−0.932, −0.287], on rows whose q is withheld (Section 2.5). A rule fixed in advance required the position gain to clear the whole of a nested null, which it did not, so transfer across participants is not established here. Correcting with a participant's own offset, a ceiling because it needs participant-specific marker data, bought 1.094° at the trunk–thigh angle and nothing at the thigh–shank angle (0.166° , $p = 0.679$). Two features of the offset may explain why. Only 11.04 mm of the knee's 32.36 mm offset lies in the plane that moves flexion (Table 1), and the offset is not constant within a participant. Across quintiles of the reference thigh–shank angle its magnitude changed by +6.19 mm at the hip, −2.82 mm at the knee and −4.27 mm at

the ankle (Fig. 2; Tables S3c, S3d). The largest posture dependence appears to be at the hip, on which the literature disagrees (Tang et al., 2022; Wade et al., 2023), although at $n = 11$ a correlation below about 0.6 would be undetectable, so neither account is refuted (Table S17).

4. Discussion

Within a participant, most of what the position metric measures at the hip and knee is a disagreement about where the joint centre lies and not an error of reconstruction. Removing the offset moved per-joint verdicts, which is the consequence for comparisons between pipeline variants. Das et al. (2023) and Pataky et al. (2013) have both shown a scalar summary over-simplifying a richer object; what this study adds is the size of that displacement against what a pipeline decision can change.

Whether a displacement of this size represents a difference of definition or an error of our own cannot be settled at the hip. The two candidate errors common to both arms, the synchronisation residual and the reference's own alignment error, are at most 15.8 mm at the hip and knee on the 95th-percentile bound and 3.62 mm respectively, against 27–37 mm. At the hip the reference centre is itself a regression from external landmarks whose published accuracy, 26–33 mm at 95% certainty, matches the offset we report (Bell et al., 1989); the knee and ankle references are marker midpoints (Tables S0, S3m). The hip offset can therefore not be apportioned between the two systems and no absolute accuracy claim follows. It is both the joint at which putting the target back is least well defined and the one carrying the largest reversal, which we cannot explain.

The offset's reproduction by three detectors in the same direction for every participant, and the outward, bilaterally symmetric mediolateral component at the hip, are both consistent with the interpretation of Needham et al. (2021), who attributed the hip and knee differences to the labelling conventions of the training data and not to the estimator, and with the

constant biases D'Haene et al. (2024) removed by subtracting a mean waveform difference before comparing systems.

This study has several limitations. Every markerless estimate was unfiltered against a 12 Hz-filtered reference, and on the one filtered subset a post-hoc test could not distinguish the angle cost from zero, which is the weaker reading we take (Table S18). The direct linear transform and Gauss–Newton comparison is confounded with a difference in objective, and the offset-removal analysis subtracts a mean offset that itself varies with posture, so what remains is not offset-free. The eleven participants are not a random eleven of the archive's fifteen, being three women short of the cohort described. Two of them shape what we report: one carries a heading span the bias-term argument of Table S14 assumes it does not have (Table S20), the other the worst case in twelve of the fourteen headline contrasts, though no contrast depends on it (Table S16). All results come from one rig, one task and one laboratory, every headline contrast from a single detector, and transverse-plane kinematics were not assessed. The family structure was also exploratory, and boundaries chosen after seeing results are a degree of freedom no correction removes. Every angle effect we report lies below the between-session minimal detectable change Wilken et al. (2012) give for sagittal knee angles, 4.16–6.39° across peaks, which may transfer as a scale but not as a threshold. Finally, a pipeline that solves inverse kinematics has no small defining point set, the analogous lever being the marker set (Mundt et al., 2025), so these results are scoped to direct triangulation and exclude the model-fitting systems of the reviews cited in Section 1.

5. Conclusion

On this nine-camera ring no per-joint triangulation effect at the hip, knee or ankle exceeded 2.5 mm and no single-step angle effect on the four chains exceeded 0.38°, as point estimates, with widest 95% limits of +3.417 mm and −0.583°. The offset correction is not a triangulation effect and moves 9.848 mm. The rejection claim rests on the narrower ± 0.35

mm at 90%, which was broken on a five-camera subset of the same ring. That ceiling is scoped to the four chains tested, since the same contrasts move the shank–foot angle by -0.522° (reprojection) and -0.643° (object-space) at the toe, where the pre-declared veto returns no verdict because the heel reverses sign (Tables S19, S21). Most of these effects sit at the sign-flip floor, where the test certifies direction and not magnitude, so we offer them as hypotheses for confirmatory work. On that basis we suggest that per-joint position verdicts be read against the offsets they contain, that position be measured on the joints the reported angle uses, and that any claimed dissociation be tested as an interaction and not as a pair of separate verdicts.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Claude (Anthropic) in order to draft and revise manuscript text, to write and run the analysis and figure-generation code, and to check the manuscript's statements against the cited sources and the result files. After using this tool the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data and code availability

The BioCV dataset is available from the University of Bath Research Data Archive (<https://doi.org/10.15125/BATH-01258>) on request under a data use agreement. All analysis

312 code — triangulation arms, permutation and interval estimation, interaction testing, and
313 figure generation — is available at **[repository DOI to be supplied on acceptance]**.

314 **Conflict of interest statement**

315 None declared.

316 **Ethics**

317 Secondary analysis of the BioCV dataset; no new data were collected and no additional
318 approval was required. The original collection was approved by the Institutional Review
319 Board of the University of Bath (EP1819052, 25/07/2019) and all participants gave written
320 informed consent to participate and to data release.

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322

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399

400 **Figure captions**

401 **Fig. 1.** On a retention control, a null in a three-joint mean is two significant effects of
402 opposite sign. 3D position effect at each lower-limb joint separately, with the three-joint
403 mean at right. At matched retention $k = 2$ (left) the object-space criterion improves the hip
404 and degrades the knee, both $q = 0.004$, cancelling to a null mean; under adaptive rejection
405 (centre) the same criterion helps the knee (0.339 mm, $q = 0.004$), and the ankle effect (0.190
406 mm) is significant unadjusted but not after correction ($q = 0.053$). Confidence weighting
407 (right) improves every joint. Bars are error against the reference as measured; filled bars
408 survive Benjamini–Hochberg correction within their family and open bars do not, which is
409 why the ankle bar is open at an unadjusted p of 0.035. What happens once the offset is
410 removed is in Section 3.2 and Table 2.

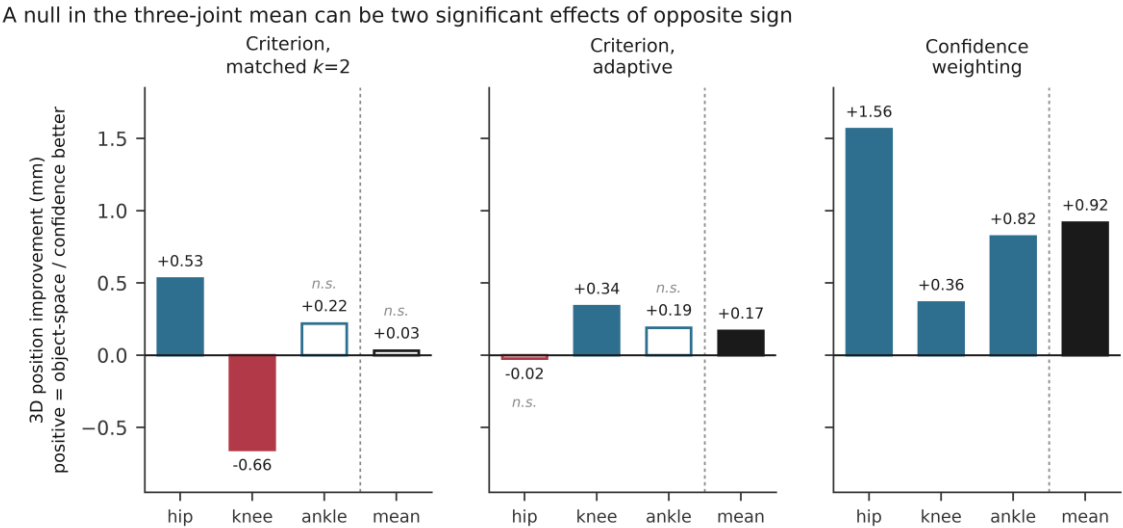


Fig. 2. The joint-centre offset is not fixed but varies with posture. Offset magnitude by quintile of the reference thigh–shank angle (180° = full extension for the three-point angle used here). Highest minus lowest quintile: hip +6.19 mm ($q = 0.0065$), knee -2.82 mm ($q = 0.029^\dagger$), ankle -4.27 mm ($q = 0.013$), each corrected within the thirteen-test posture family; † marks the one that does not survive the arbitrary-dependence sensitivity of Table S5.

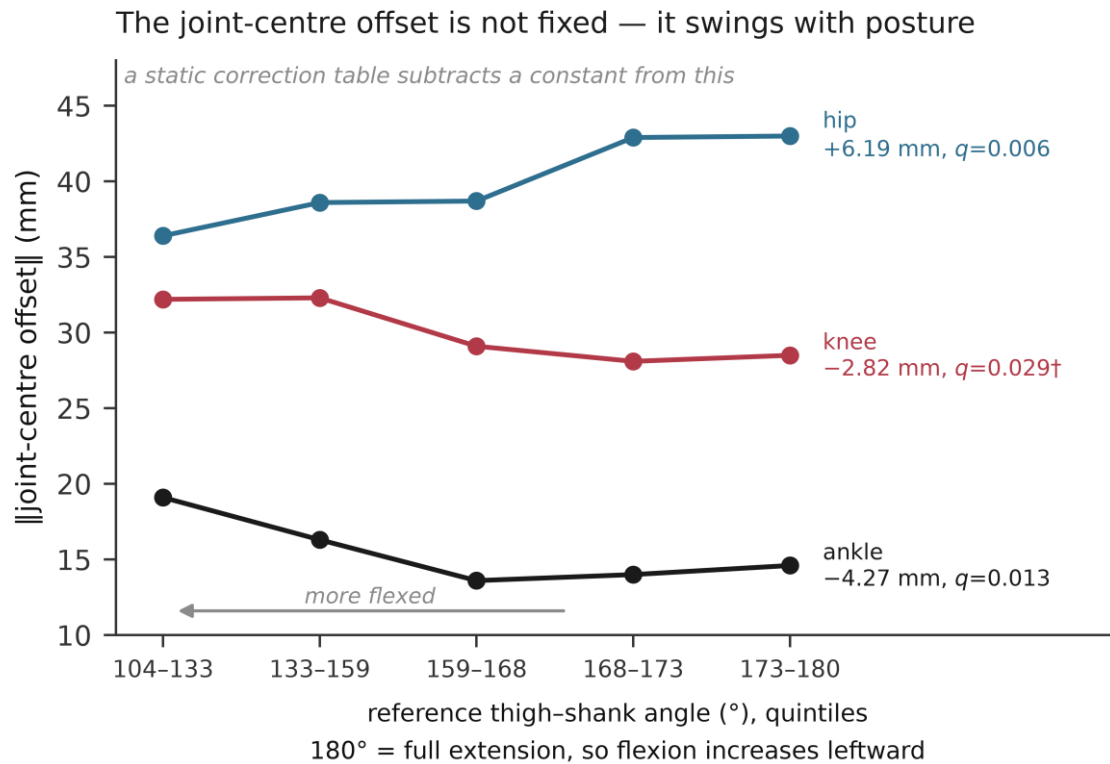


Fig. 3. (a) Thigh–shank angle error against the number of cameras discarded per joint at matched retention, tested per participant against $k = 0$; all six steps survive correction within the dose-response family. Matched retention is a control: it fixes the number discarded at levels the adaptive rule never reaches, so the rate carries no deployable claim. The panel quotes the tested mean rate, not the marginal increment of Table 3(b). (b) The object-space advantage in peripheral-set position by camera-distance asymmetry, before and after lens-distortion correction (§3.3); the same control on the lower-limb set is in Table S2d. Filled bars have bootstrap intervals excluding zero.

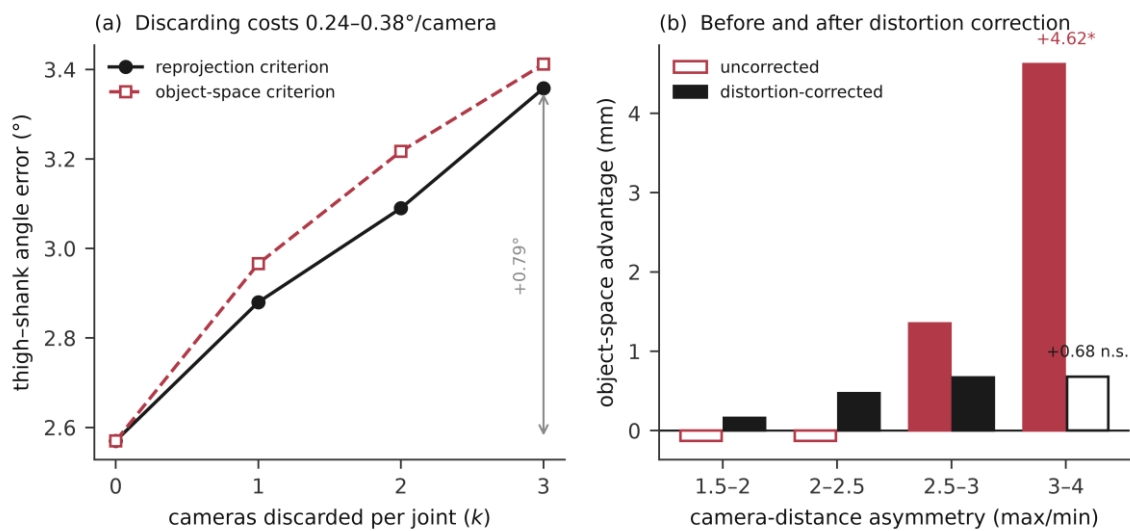


Fig. 4. A position gain and an angle change need not travel together, and the gap is what is tested. For each intervention the open marker is the 3D position effect on hip, knee and ankle and the filled marker is the joint-angle effect, both as a percentage of that arm's baseline error; the line is the gap between them and the right-hand number is the interaction (position minus angle, percentage points). Nine of the twenty declared interactions are drawn: every intervention that appears in a headline claim, at the angle that claim is made on. An asterisk marks $q < 0.05$ within the family of twenty; the offset-correction rows carry ‡ and no q , because Section 2.5 withholds it on leave-one-out rows. (a) Triangulation interventions. (b) The joint-centre offset correction, on its own scale.



Tables

Table 1. The joint-centre offset, and why correcting it fails

(a) Correcting the offset (mean absolute error, degrees)

angle	uncorrected	population (leave-one-out) correction	own-offset ceiling
thigh–shank angle	2.575	3.201	2.495
frontal-plane thigh–shank	2.637	2.768	2.486
trunk–thigh angle	3.651	3.363	2.541

The deployable correction worsens the thigh–shank angle by 0.606° , on a row whose q is withheld ($\ddagger$). The own-offset column is an upper bound requiring participant-specific marker data, and buys only 0.080° at the thigh–shank angle against 1.110° at the trunk–thigh angle.

The column arithmetic does not equal the tested effect, and should not. Differencing this table gives 0.626° at the thigh–shank angle, 0.131° at its frontal-plane form and 0.288° at the trunk–thigh angle, against tested effects of -0.606° , -0.149° and $+0.164^\circ$. The table reports **means pooled over all observations**; the tested effect is the **mean of the per-participant differences**, which is the inferential unit throughout (Section 2.5). The two coincide only when every participant contributes equally, and participants here contribute between 6 and 10 usable trials (P26 lost four; P04 was archived with nine and P28 lost one trial to missing reference data). Unequal weighting accounts for the gap to within rounding: computing the same three contrasts by pooling observations reproduces the column arithmetic (-0.626 , -0.130 , $+0.288^\circ$), and computing them per participant reproduces the tested effects (-0.606 , -0.149 , $+0.164^\circ$). The own-offset column reconciles closely at the hip, 1.110° pooled against 1.094° tested, and not at the thigh–shank angle, 0.080° against 0.166° , a factor of two. Where the two differ we quote the per-participant value, since that is what was tested.

(a2) How much of the position error is a systematic offset — bias/variance split of the 3D error vector, world frame, on the adaptive object-space-rejection arm with confidence

weights. The offset is estimated WITHIN each participant and then averaged across them, so between-participant variation in the offset does not enter the random term; the body-frame split of Table S3m differs by less than 0.5 mm

joint	RMSE (mm)	systematic offset (mm)	random residual (mm)	offset share of squared error
hip	41.2–41.3	35.4–36.9	18.6–21.1	74–80%
knee	31.0–33.4	27.5–30.1	14.3–14.6	79–81%
ankle	17.3–17.4	11.5	13.0	44%

Pooled across participants, which is what a cohort-level validation study measures, the between-participant scatter of the offset (hip 20.0–21.9 mm, knee 11.2–13.4 mm, ankle 6.2–6.7 mm) moves out of the offset term and into the random one. The share is then **47–57%** at the hip, **61–70%** at the knee and **30–32%** at the ankle, in the world frame declared above. Both columns are reported because they answer different questions: this paper's contrasts are differences within a participant, where that participant's own offset is common to both arms, so the within-participant share is the one they rest on.

Every range in this panel is the span between the left and right sides, not a confidence interval. The per-joint effects contrasted in Table 4(b) span 0.024 to 1.564 mm. Decomposing the estimator change as well gives 2.415 mm at the hip for DLT to Gauss-Newton and 2.469 mm for adding rejection under it (Table S8), so the largest per-joint triangulation effect at the hip, knee and ankle is 2.5 mm, one order of magnitude below the offset at hip and knee.

(b) Why the thigh–shank angle gains so little — knee offset decomposed relative to the hip–ankle chord

component	mean magnitude (mm)	effect on flexion
axial, along the limb	27.45	changes limb length, not flexion
in-plane perpendicular	11.04	drives flexion error
out-of-plane	8.63	affects ab/adduction, not flexion
total loffsetl	32.36	—

Each entry is a mean of per-frame absolute values, so the components do not combine into the total by root sum of squares; the total is the mean of the per-frame offset magnitude. That is also why 32.36 mm exceeds the 27.5–30.1 mm systematic knee offset of Table 1(a2), which is the norm of the mean: a mean of magnitudes carries the per-frame scatter that a norm of a mean removes. The two tables do not disagree.

(c) The offset varies with posture — binned by quintile of the ground-truth thigh–shank angle; highest minus lowest quintile, per participant, exact sign-flip

joint	loffsetl across quintiles (mm)	change	<i>q</i>
hip	36.4 → 38.6 → 38.7 → 42.9 → 43.0	+6.19	0.0065
knee	32.2 → 32.3 → 29.1 → 28.1 → 28.5	−2.82	0.029†
ankle	19.1 → 16.3 → 13.6 → 14.0 → 14.6	−4.27	0.013

The quintile columns are means pooled over observations; the change column is the mean of the per-participant differences, which is the tested quantity (§2.5). The two differ because participants contribute unequal numbers of trials, exactly as set out under Table 1(a).

Individual components move further than the magnitudes, because direction rotates as well as length: the hip's anterior component by −13.60 mm ($p = 0.001$) and the knee's superior component by −8.97 mm ($p = 0.001$).

494 **Table 2. The per-joint contrasts repeated on offset-removed error**

495 Each participant's own mean error vector for a joint is subtracted from both arms of a contrast,
496 so what remains is the perturbation and not the displaced target. Subtracting one arm's own
497 centroid centres that arm and not the other, which biases the contrast downward when the
498 offset comes from the first-named arm and upward when it comes from the second. The
499 midpoint of the two is symmetric and carries every verdict; the single-arm range is the span
500 of that bias and is shown for reference only — a range must not be built from it. *As measured*
501 repeats Table 4(b) for the first four contrasts; the last two are the rejection contrasts the
502 Conclusion's position null rests on, added under a rule fixed before they were run.

503 Effects are millimetres, positive meaning the second-named arm is better. p is the exact sign-
504 flip test over all 2^{11} sign patterns; q is Benjamini–Hochberg within $m = 18$, run once per
505 reference column so that each column is a complete correction over the same eighteen
506 contrasts. Bold marks a midpoint effect surviving that correction, and † one that does not
507 survive the arbitrary-dependence sensitivity of Table S5 — the same convention as Tables 1
508 and 4, applied here so the absence of daggers is not read as a property of the panel. No row
509 has a midpoint verdict differing from both single-arm verdicts. The component
510 decomposition, the single-arm columns in full, the per-subset control and the
511 dispersion/centroid split are in Table S7.

contrast	joint	as measured	p	offset-removed	p	q	single-arm range
adaptive criterion	hip	-0.024	0.8350	+0.264	0.0010	0.0035	[+0.221, +0.308]
adaptive criterion	knee	+0.339	0.0020	+0.181	0.0547	0.1094	[+0.170, +0.192]
adaptive criterion	ankle	+0.190	0.0352	+0.150	0.1084	0.1774	[+0.142, +0.158]
criterion at matched k=1	hip	+0.253	0.0840	-0.718	0.0010	0.0035	[-0.800, -0.632]
criterion at matched k=1	knee	-0.322	0.0029	+0.399	0.0039	0.0100	[+0.372, +0.426]
criterion at matched k=1	ankle	+0.134	0.1484	+0.076	0.3066	0.3943	[+0.063, +0.088]
criterion at matched k=2	hip	+0.530	0.0020	-0.108	0.5771	0.6790	[-0.222, +0.009]
criterion at matched k=2	knee	-0.655	0.0020	+0.738	0.0029	0.0088	[+0.643, +0.828]
criterion at matched k=2	ankle	+0.219	0.0762	+0.215	0.0664	0.1195	[+0.181, +0.247]
uniform $\rightarrow$ confidence	hip	+1.564	0.0010	+0.253[†]	0.0088	0.0198	[+0.069, +0.435]
uniform $\rightarrow$ confidence	knee	+0.365	0.0010	-0.024	0.7266	0.7693	[-0.031, -0.017]
uniform $\rightarrow$ confidence	ankle	+0.822	0.0010	+0.912	0.0010	0.0035	[+0.903, +0.921]
no rejection $\rightarrow$ object-space rejection	hip	+0.608	0.0234	-1.726	0.0010	0.0035	[-1.878, -1.564]
no rejection $\rightarrow$ object-space rejection	knee	-0.546	0.0039	-0.020	0.8486	0.8486	[-0.086, +0.046]
no rejection $\rightarrow$ object-space rejection	ankle	+0.426	0.0039	+0.256	0.1621	0.2432	[+0.213, +0.299]
no rejection $\rightarrow$ reprojection rejection	hip	+0.633	0.0439	-1.983	0.0010	0.0035	[-2.240, -1.714]
no rejection $\rightarrow$ reprojection rejection	knee	-0.885	0.0010	-0.203	0.2051	0.2840	[-0.279, -0.125]
no rejection $\rightarrow$ reprojection rejection	ankle	+0.237	0.1738	+0.107	0.6035	0.6790	[+0.084, +0.130]

What the table shows. At the hip the adaptive criterion and both rejection contrasts change sign once the target is put back, and the two rejection reversals are the ones Section 3.2 reports; their as-measured values are uncorrected and outside the declared per-joint family. At the knee the two matched- k contrasts reverse with both directions surviving correction. The ankle moves nowhere.

Table 3. Joint angles

(a) Contrasts (degrees; positive = the second-named arm has lower error)

contrast	dep.	thigh–shank angle	frontal-plane thigh–shank	trunk–thigh angle
uniform vs confidence weighting	y	+0.023, n.s.	+0.078, q = 0.004	+0.227, q = 0.006
no rejection vs reprojection rejection, adaptive	y	−0.188, q = 0.021†	−0.013, n.s.	−0.192, q = 0.022†
no rejection vs object-space rejection, adaptive	y	−0.141, q = 0.011	−0.067, raw only	−0.167, q = 0.028†
reprojection vs object-space criterion, adaptive	y	+0.047, n.s.	−0.054, n.s.	+0.025, n.s.
confidence vs confidence $\times 1/d$ weighting, no rejection	n	+0.052, n.s.	−0.188, q = 0.004	−0.219, q = 0.004
reprojection vs object-space criterion, matched $k = 1$	n	−0.089, q = 0.016†	—	—
reprojection vs object-space criterion, matched $k = 2$	n	−0.129, q = 0.013†	—	—
no rejection vs reprojection rejection, under Gauss–Newton	y	−0.169, q = 0.028†	—	—
uncorrected vs leave-one-participant-out offset correction	y*‡	−0.606, ‡ withheld	−0.149, ‡ withheld	+0.164, ‡ withheld

dep. marks an arm a validation study could deploy; **n** marks a diagnostic or control arm — the distance weightings use ground-truth distances, and matched- k fixes retention at a level the adaptive rule never reaches. **y*** marks the population offset table, which is deployable in principle but is computed here as a world-frame translation on trials whose headings span 56° across the study and 1.0 – 14.1° within ten of the eleven participants, so this particular table does not transfer to another laboratory or protocol (Table S20); its dissociation is demonstrated within this rig. † marks a verdict that does not survive the arbitrary-dependence

sensitivity of Table S5, i.e. one resting on the positive-dependence assumption. Exact 95% intervals for every surviving row are in Table S8.

(b) Dose–response, thigh–shank angle — mean absolute error pooled over observations (degrees), and the same steps tested per participant against $k = 0$, BH-corrected within a family of 6

cameras discarded per joint	0	1	2	3
reprojection criterion, pooled MAE	2.570	2.880	3.090	3.358
object-space criterion, pooled MAE	2.570	2.966	3.217	3.412
reprojection, tested cost (q)	—	+0.286 (0.011)	+0.482 (0.006)	+0.747 (0.006)
object-space, tested cost (q)	—	+0.375 (0.006)	+0.612 (0.006)	+0.798 (0.006)

The $k = 0$ baseline here is 2.570° where Table 1(a) and Table S2c give 2.575° for the same arm. The dose-response family is accumulated over the 12,468 observations on which every retention level is defined, a subset of the frames the main family uses, so the two baselines are the same quantity on different frame sets and are not reconciled to a single value.

All six steps survive correction. The tested cost averages **$0.24\text{--}0.38^\circ$ per camera discarded**, that is the cumulative cost at k divided by k ; the per-participant $k = 0$ baseline is 2.669° , against the 2.570° pooled mean above, for the reason set out under Table 1(a). Only the tested values are quoted in the text. Differencing the pooled row gives a MARGINAL increment of $0.20\text{--}0.40^\circ$ for each further camera, which is a different quantity from the mean rate above and is arithmetic on means rather than the contrast that was tested.

544 **(c) Dissociation tested as an interaction** — per participant, relative position effect minus
 545 relative angle effect, on one dimensionless scale; BH-corrected within the family of 20

intervention	dep.	Δ position	Δ angle	interaction	q
criterion at matched $k = 1$, thigh–shank angle	n	+0.07%	−3.21%	+3.29	0.005
confidence->confidence x 1/d, trunk–thigh angle	n	−2.46%	−6.58%	+4.12	0.005
LOO offset correction, thigh–shank angle	y*‡	+34.04%	−27.03%	+61.07	‡ withheld
LOO offset correction, frontal-plane thigh–shank	y*‡	+34.04%	−10.31%	+44.35	‡ withheld
confidence->confidence x 1/d, frontal-plane thigh–shank	n	−2.46%	−7.67%	+5.22	0.008
LOO offset correction, trunk–thigh angle	y*‡	+34.04%	−4.48%	+38.52	‡ withheld
criterion at matched $k = 2$, thigh–shank angle	n	+0.13%	−4.01%	+4.14	0.010
Gauss–Newton with rejection (w^2 -matched), thigh–shank angle	y	+3.45%	−7.45%	+10.90	0.010
no rejection → object-space rejection, thigh–shank angle	y	+0.51%	−6.09%	+6.60	0.012
GN: no rejection → reproj rejection, thigh–shank angle	y	+2.79%	−7.25%	+10.04	0.012
no rejection → object-space rejection, trunk–thigh angle	y	+0.51%	−5.83%	+6.34	0.014 †
no rejection → reprojection rejection, thigh–shank angle	y	−0.10%	−8.16%	+8.07	0.024 †
reprojection → object-space criterion, frontal-plane thigh–shank	y	+0.61%	−2.61%	+3.22	0.035 †
uniform → confidence weighting, trunk–thigh angle	y	+3.10%	+5.69%	−2.59	0.071
DLT → Gauss–Newton (w^2 -matched), thigh–shank angle	y	−3.31%	+0.22%	−3.54	0.215
criterion at matched $k = 3$, thigh–shank angle	n	+0.58%	−1.74%	+2.31	0.238
uniform → confidence weighting, thigh–shank angle	y	+3.10%	+1.43%	+1.67	0.284
reprojection → object-space criterion, thigh–shank angle	y	+0.61%	+1.74%	−1.13	0.377
GN objective: w^2 -matched → unmatched (w), thigh–shank angle	y	−1.53%	−1.03%	−0.50	0.409
uniform → confidence weighting, frontal-plane thigh–shank	y	+3.10%	+3.06%	+0.04	0.947

546

547 A position-against-angle interaction cannot be run in native units, the two outcomes having
 548 different dimensions, so §2.5's both-scales rule is unenforceable for this family and every row
 549 here is relative-scale only. Where a row's two components share a sign its magnitude is
 550 denominator-dependent; where they oppose, the sign is not. All 20 interactions in the family
 551 are listed, survivors in bold. Thirteen of the twenty survive correction as the artefact prints it,
 552 but three of those are offset-table rows whose q is withheld (‡), so **ten interactions carry a**
 553 **valid verdict**, six of them on deployable arms; 1 that was significant unadjusted does not
 554 survive. Only ONE row pairs two individually surviving effects of opposite sign: an offset-
 555 table row cannot, because under the rule of Section 2.5 neither of its components has a valid

556 q. That one row, Gauss–Newton rejection, itself survives the arbitrary-dependence sensitivity:
557 it sits at rank 10 of 20 with $p = 0.0059$ against a Benjamini–Yekutieli threshold of 0.00695
558 there ($m = 20$, $H_m = 3.60$), and ten of the twenty survive. Its angle component carries a
559 dagger. Section 3.3 states it that way. *dep.* marks an arm a validation study could deploy; **n**
560 marks a diagnostic or control arm — the confidence $\times 1/d$ rows use ground-truth distances
561 and the matched- k rows fix retention at a level the adaptive rule never reaches. ‡ marks a row
562 on which no q is reported. Each participant's correction is built from the other ten, so the
563 eleven differences are not exchangeable under the null and the sign-flip test does not apply; a
564 rule declared before those rows were run required the q to be withheld. The only inferential
565 statement available for them is a nested null on the [pos:HKA] gain, one-sided $p = 0.026$
566 (§2.5, Table S3e). **y*** marks the population offset table, deployable in principle but computed
567 here as a world-frame translation on trials whose headings span 56° across the study and 1.0 –
568 14.1° within ten of the eleven participants, so this table does not transfer beyond this rig
569 (Table S20). † marks a verdict that does not survive the arbitrary-dependence sensitivity of
570 Table S5. Exact 95% intervals for every surviving row are in Table S8. The 7 that do not are
571 shown so that the family can be audited rather than summarised.

A surviving interaction is not by itself a dissociation. Two rows pair component effects of opposite sign — LOO offset correction at the thigh–shank angle, and Gauss–Newton no rejection → reprojection rejection at the thigh–shank angle — but only the second has a valid q on BOTH components, because under the rule of Section 2.5 neither component of an offset-table row has one. One row therefore pairs two individually SURVIVING effects of opposite sign, and Section 3.3 reports that one. Elsewhere the two components share a sign (the confidence $\times 1/d$ rows, which harm both outcomes unequally) or one component is null on its own (the matched- k and no-rejection rows, whose position half does not move). Read each row's two component columns before reading its interaction.

Scale. Δ position and Δ angle are **means of per-participant ratios** — each participant's effect divided by that participant's own baseline error, then averaged — not ratios of pooled means. The two differ only slightly once both use the baseline arm as denominator: the LOO position effect is +34.04% here against +9.848 mm on a 28.5 mm pooled baseline (34.6%). The per-participant form is used because the interaction is a within-participant contrast and the inferential unit is the participant.

588 **Table 4. 3D position error: the joint set changes the answer**

589 **(a) The same contrasts on two joint sets (mm)**

contrast	dep.	ankles + wrists	hip, knee and ankle
uniform vs confidence weighting	y	+5.062 [+4.309, +5.930], q = 0.004	+0.917, q = 0.004
confidence vs confidence $\times 1/d$ weighting, no rejection	n	+0.256, q = 0.038†	-0.691, q = 0.004
reprojection vs object-space criterion, adaptive	y	+0.332, q = 0.004	+0.168, q = 0.018†
reprojection vs object-space criterion, matched $k = 1$	n	+0.148, q = 0.009	+0.029, n.s.
reprojection vs object-space criterion, matched $k = 2$	n	+0.428, q = 0.004	+0.039, n.s.
reprojection vs object-space criterion, matched $k = 3$	n	+0.782, q = 0.004	+0.169, n.s.
no rejection vs reprojection rejection, adaptive	y	+0.131, n.s.	-0.005, n.s.
no rejection vs object-space rejection, adaptive	y	+0.463, n.s.	+0.163, n.s.
uncorrected vs leave-one-participant-out offset correction	y*‡	—	+9.848, ‡ withheld
DLT vs Gauss–Newton, no rejection	y	—	-1.384, q = 0.004
Gauss–Newton: no rejection vs reprojection rejection	y	—	+0.858, q = 0.006

590

591 Bold marks a multiplicity-surviving effect in the hip/knee/ankle column, which is where the
592 paper's claims about the joints an angle uses are made. *dep.* marks an arm a validation study
593 could deploy; **n** marks a diagnostic or control arm that cannot carry a practical claim — the
594 two distance weightings use ground-truth distances, and matched- k fixes retention at a level
595 the adaptive rule never reaches. ‡ marks a row on which no q is reported. Each participant's
596 correction is built from the other ten, so the eleven differences are not exchangeable under the
597 null and the sign-flip test does not apply; a rule declared before those rows were run required
598 the q to be withheld. The only inferential statement available for them is a nested null on the
599 [pos:HKA] gain, one-sided $p = 0.026$ (§2.5, Table S3e). **y*** marks the population offset table,
600 deployable in principle but computed here as a world-frame translation on trials whose
601 headings span 56° across the study and 1.0 – 14.1° within ten of the eleven participants, so this
602 table does not transfer beyond this rig (Table S20). † marks a verdict that does not survive the
603 arbitrary-dependence sensitivity of Table S5. Exact 95% intervals for every surviving row are
604 in Table S8.

Each row's baseline is the arm it names first; for the no-rejection confidence arm that is 16.6 mm (ankles + wrists) and 28.5 mm (hip, knee, ankle), and the percentages in Table 4(c) use each row's own baseline rather than these two. The hips dominate the latter at 37.7–39.3 mm. Confidence $\times 1/d$ weighting reverses sign between the two joint sets, and the matched- k criterion advantage disappears entirely.

(b) Decomposed by joint (mm; exact sign-flip p , uncorrected, within this analysis; bold marks $p < 0.05$ among the per-joint entries, not a corrected verdict. The per-joint q -values are in Table S3b, where seven of the twelve survive and the bold ankle entry in the adaptive row does not; exact intervals are in Table S8)

contrast	hip	knee	ankle	three-joint mean
reprojection vs object-space, matched $k = 1$	+0.253 (0.084)	− 0.322 (0.003)	+0.134 (0.148)	+0.022 (0.706)
reprojection vs object-space, matched $k = 2$	+0.530 (0.002)	− 0.655 (0.002)	+0.219 (0.076)	+0.031 (0.654)
reprojection vs object-space, adaptive	−0.024 (0.835)	+0.339 (0.002)	+0.190 (0.035)	+0.168 (0.008)
uniform $\rightarrow$ confidence weighting	+1.564 (0.001)	+0.365 (0.001)	+0.822 (0.001)	+0.917 (0.001)

At matched retention the criterion improves the hip and degrades the knee, both significantly, and the three-joint mean cancels them to a null. Confidence weighting improves every joint. The mean column is the unweighted mean of the three per-joint effects and belongs to the within-limb per-subset family ($m = 20$), not to the $m = 50$ family of panel (a): the two differ by ≤ 0.01 mm because panel (a) is computed over a 0.1% different observation set. Table S7 repeats every row here with the joint-centre offset removed.

(c) Does the joint set change the effect? — per participant, the effect on ankles and wrists minus the effect on hip/knee/ankle, each as a percentage of that participant's own baseline at that joint set; BH-corrected within the family of 8

contrast	Δ [pos:AW]	Δ [pos:HKA]	interaction	q
uniform vs confidence weighting	+23.41%	+3.10%	+20.31	0.002
criterion at matched $k = 2$	+2.54%	+0.13%	+2.41	0.002
criterion at matched $k = 3$	+4.49%	+0.58%	+3.91	0.002
confidence vs confidence $\times 1/d$ weighting	+1.46%	−2.46%	+3.91	0.002
criterion, adaptive rejection	+1.95%	+0.61%	+1.34	0.005
criterion at matched $k = 1$	+0.92%	+0.07%	+0.84	0.014
no rejection vs object-space rejection	+2.48%	+0.51%	+1.97	0.172
no rejection vs reprojection rejection	+0.54%	−0.10%	+0.64	0.618

All 8 are listed. 6 survive correction: all three matched-retention contrasts, which are controls; the confidence $\times 1/d$ contrast that carries the sign reversal; and two deployable ones, uniform versus confidence weighting ($q = 0.002$) and the adaptive criterion ($q = 0.005$). The 2 that do not are the rejection-versus-none contrasts, which are null on both joint sets. This interaction is run on the relative scale only, so by Section 2.5's rule it is scale-dependent; the within-limb control of panel (d), run on both scales, carries the claim.

(d) Within one limb: only the membership of the set changes. Eighty contrasts across three declared families, all corrected and all reported in Table S3n. They are not reproduced here: the thirty pairwise differences on each scale span ten free quantities rather than thirty, and after both declared sensitivities the panel reduces to a single deployable contrast — confidence weighting, {hip, knee, ankle} against {hip, ankle}, −1.05% and −0.276 mm ($q = 0.004$) — which Section 3.2 quotes in full. The families remain in the multiplicity accounting of Table S4b; only their presentation moves.